



Pre- and postpartum metabolizable protein supply: II. Effects on plasma amino acids and markers of tissue mobilization in transition Holstein dairy cows.

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ABSTRACT

The influence of diet composition on the degree of adipose and lean muscle mobilization and concentrations of circulating AA has been demonstrated during the transition period. Altering the MP supply might offer a strategy to control tissue mobilization and increase circulating AA availability, but the optimum supply of MP fed pre- and postpartum remains unknown. We investigated the effect of increasing the MP supply in the prepartum, postpartum, or both diets on plasma AA concentrations and ultrasound and circulating indicators of tissue mobilization. Multiparous Holstein cows ($n = 96$) were assigned to 1 of 4 treatment groups at 28 d before expected calving following a randomized block design. Prepartum diets were formulated to contain either a control (C; 85 g of MP/kg DM; 1,175 g of MP/d) or high (H; 113 g of MP/kg DM; 1,603 g of MP/d) level of estimated MP. From calving to 21 DIM, fresh diets were formulated to contain either a control (C; 104 g of MP/kg DM; 2,044 g of MP/d) or high (H; 131 g of MP/kg DM; 2,685 g of MP/d) level of estimated MP. To control the potential confounding effect of Met and Lys supply, diets were formulated to supply an equal amount at 1.24 and 3.84 g/Mcal of ME in both prepartum diets and 1.15 and 3.16 g/Mcal of ME in both postpartum diets, respectively. The combination of a pre- and postpartum diet resulted in 4 treatment groups: 1) CC ($n = 23$), 2) CH ($n = 24$), 3) HC ($n = 22$), and 4) HH ($n = 23$). A common lactation diet (113 g of MP/kg DM; 2,956 g of MP/d) was fed from 22 DIM to the end of the observation period at 42 DIM. Transcutaneous ultrasonography was used to determine the longissimus dorsi muscle diameter and backfat thickness. Concentrations of plasma AA, 3-methylhistidine (3MH), and creatinine were determined on a subset of cows ($n = 60$) using ultra-high-performance liquid chro-

matography and mass spectrometry. Treatment did not affect the longissimus dorsi muscle diameter from -14 to 21 d relative to calving, but the diameter was greater in CH compared with HH at 40 DIM. Backfat thickness and the ratio of 3MH to creatinine did not differ by treatment. Concentrations of EAA were greater at -13 d relative to calving in HH compared with CC and CH and at -6 d relative to calving EAA concentrations were higher in HC compared with CC. Cows fed the H diet postpartum had elevated EAA concentrations at 6 and 20 DIM compared with cows fed the C postpartum diet but EAA concentration did not differ at 40 DIM. Total NEAA concentrations were higher in CH compared with HC and HH at -6 d relative to calving, but NEAA concentration did not differ by treatment at -13, 6, 20, or 40 d relative to calving. In conclusion, increasing the supply of MP fed prepartum, postpartum, or both had minimal effects on tissue mobilization but influenced concentrations of plasma AA.

Key Words: Metabolizable protein, transition cow, plasma amino acids, tissue mobilization

INTRODUCTION

Transition dairy cows are challenged by inadequate intake to meet their nutrient demands and subsequently enter a state of negative energy and protein balance (Drackley, 1999; Mann et al., 2015; Mann et al., 2016a). As part of a normal homeorhetic adaptation during this time of negative energy and protein balance, cows mobilize muscle and adipose tissue (McCabe and Boerman, 2020; Mann, 2022). While some degree of tissue mobilization appears to be a hallmark of the adaptation to lactation in high-producing dairy cows, excess mobilization has been associated with negative production outcomes and health events (Ospina et al., 2010; Siachos et al., 2022; Peiter et al., 2023). Van der Drift et al. (2012) and Bell et al. (2000) suggest that protein mobilization can begin prepartum and increase to as much as 1,000 g/d by 7 to 10 DIM. However, diet formulation has been shown to

Received April 8, 2024.

Accepted August 18, 2024.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-24. Nonstandard abbreviations are available in the Notes.

influence the timing and extent of mobilization (Doepel et al., 2002; Mann et al., 2016a; Zang et al., 2021) and altering dietary MP supply during the transition period might offer a strategy to improve circulating AA availability and control the extent of tissue mobilization.

Using 3-methylhistidine (3MH) as an indicator of muscle breakdown, Doepel et al. (2002) showed prepartum protein mobilization was lower when feeding a close-up diet containing 17.0 compared with 12.5% CP. As such, feeding greater than 1,000 g of MP/d has been recommended in the close-up period to preserve labile protein reserves during gestation (Ji and Dann, 2013; Van Saun and Sniffen, 2016; Nydam et al., 2017). When more MP was supplied in early lactation, concentrations of 3MH were lower (Carder and Weiss, 2017; Tebbe and Weiss, 2021; Zang et al., 2022), and concentrations of plasma EAA were higher (Tebbe and Weiss, 2021). Further, when infusing casein into the abomasum until 29 DIM at a rate designed to eliminate the protein deficit, circulating total AA as well as concentrations of fatty acids were greater at 4 DIM compared with control animals, a situation indicative of greater adipose tissue mobilization in response to casein infusion (Larsen et al., 2014). Although MP supply has been shown to alter early lactation milk production (Amanlou et al., 2017; Farahani et al., 2017; Zang et al., 2022), the timing and extent of tissue mobilization when cows are supplemented with MP during the prepartum, postpartum, or both periods remains unknown. Moreover, the effect of pre- and postpartum MP supplementation on concentrations of plasma AA merit further investigation. As such, we designed an experiment to examine tissue mobilization as well as concentrations of plasma AA from cows fed one of 4 feeding strategies, differing in MP supply, during the transition period. We hypothesized that increasing the pre- and postpartum MP supply would synergistically alter markers of tissue mobilization and concentrations of plasma AA in transition dairy cows. The objective of our study was to compare the effects of 4 periparturient feeding strategies, differing in MP supply, on markers of tissue mobilization and plasma AA concentrations.

MATERIALS AND METHODS

All procedures were approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2019–0031). Animals, diets, and management were described in detail in our companion paper (Westhoff et al., 2024). Briefly, multiparous Holstein cows ($n = 96$) were enrolled in a randomized block design between May and November 2021. Cows were moved from a far-off dry cow pen to individual tie stalls between 35 and 42 d before expected parturition and fed a far-off diet. At 28 d before expected calving, animals were

blocked by calving date and randomly assigned within block to 1 of 4 dietary treatments groups consisting of a combination of a pre- and postpartum diet. The close-up diets were formulated to contain either a control (C; 85 g of MP/kg DM) or high (H; 113 g of MP/kg DM) level of MP. Rumen protected Met (Smartamine M, Adisseo Inc.) and Lys (USA Lysine, Kemin Industries Inc.) were used to formulate both close-up diets to supply 1.24 and 3.84 g/Mcal of metabolizable energy (ME), respectively. From calving to 21 DIM, cows were fed a fresh TMR formulated to contain either a control (C; 104 g of MP/kg DM) or high (H; 131 g of MP/kg DM) level of MP. Fresh diets were both formulated to supply Met and Lys at 1.15 and 3.16 g/Mcal of ME, respectively. The combination of a pre- and postpartum diet resulted in 4 treatment groups: 1) control-control (CC), 2) control-high (CH), 3) high-control (HC), and 4) high-high (HH), respectively. From 22 to 42 DIM cows were fed a common lactation diet, except 30 cows (CC: $n = 7$; CH: $n = 8$; HC: $n = 7$; HH: $n = 8$) that remained on their fresh treatment diet to enter a separate study objective (data not shown) and were fed the common lactation diet with approximately 5 d delay, starting at 27.2 ± 2.1 DIM. Cows were fed once daily starting at 0630 h and offered feed was adjusted to allow a minimum of 5% refusals.

Diet formulation, diet analyses, and model estimations are described in detail in Westhoff et al. (2024). In brief, diets were formulated in the Cornell Net Carbohydrate and Protein System v. 6.5.5 (CNCPS; AMTS.Cattle.Professional v. 4.17.0.0; AMTS LLC; Van Amburgh et al., 2015). The close-up and fresh C diets were formulated to meet the CNCPS MP requirement during the close-up period and represent MP concentrations fed during the fresh period on commercial dairy farms (Kerwin et al., 2022). The targeted MP concentrations of the close-up and fresh H diets were established to exceed the estimated CNCPS MP requirement during the prepartum period and to decrease the MP deficit during the fresh period, respectively. Because Met and Lys have been established as first-limiting AA in dairy cattle, treatment diets were balanced for Met and Lys to control for the confounding effect of these AA on the responses investigated in this study. Monthly composite samples of the TMR were submitted to a commercial laboratory (Cumberland Valley Analytical Services) for prediction of chemical composition by near-infrared reflectance spectroscopy and wet chemistry analysis of minerals. Wet chemistry and NDF digestibility were performed on samples of each forage composited for the length of the study. The balance of MP and ME supplied by the diet were estimated in AMTS using the results of the forage analysis as well as the group average DMI, BW, BCS, calf birth weight, and days carried calf for prepartum estimates and group average DMI, BW, BCS, DIM, milk yield, and milk composition

for postpartum estimates. To describe the contribution of tissue mobilization, determined by BCS change, on MP and ME balances as well as estimate nitrogen excretion, estimates were performed again, as described above, and included the change in BCS during the prepartum period as well as from 1 to 3 and 4 to 6 wk in milk.

Ultrasound Measurements

Once before treatment assignment (−30 d before expected calving) and on −14, −7, 7, 21, and 40 d relative to calving, the longissimus dorsi muscle diameter and subcutaneous backfat thickness (BFT) were determined by ultrasound according to the methods previously described in detail (Mann et al., 2016a; Mann et al., 2016b). Briefly, all measurements were performed on the right side of the animal using a 5–9 MHz broadband linear transducer on a portable ultrasound (IBEX PRO; E.I. Medical Imaging). The skin in the loin and thurl region was brushed and 70% alcohol was applied as a coupling agent. Longissimus dorsi diameter was measured perpendicular to the spine as the largest diameter between muscular fascial layers at the fourth transverse process. Subcutaneous backfat thickness was measured approximately 10 cm caudal of the tuber coxae as the distance between the profound fascia above the gluteus medius and the surface, excluding the measurement of the skin. The hair at the location of the ultrasound transducer during the first time point was clipped to ensure repeated placement at subsequent time points. Each measurement was performed, avoiding pressure, in triplicate and averaged for analysis.

Plasma Amino Acids

Plasma AA were determined once before treatment assignment (−30 d before expected calving) and on −13, −6, 6, 20, and 40 d relative to calving on a random subset of cows from each treatment ($n = 15$) using ultra-high-performance liquid chromatography (LC) and mass spectrometry (MS) according to the methods previously described in detail (Ortega and Van Amburgh, 2024). In brief, blood was collected before feed delivery (0600 to 0630 h) from the coccygeal vein or artery into evacuated collection tubes (Becton Dickinson) containing 158 USP units of sodium heparin. Plasma was harvested by centrifugation at 2,300 $\times$ g at 4°C for 20 min and subsequently deproteinized by diluting plasma 2-fold with a commercial solution (Seraprep, Pickering Laboratories). Samples were vortexed and stored at 4°C for 24 h before being frozen and stored at −80°C until analysis. Samples were thawed at room temperature and thoroughly vortexed before supernatants were collected by centrifugation at 15,700 $\times$ g at 4°C for 10 min. Super-

natants were diluted 10-fold with 0.1% formic acid in 50:50 acetonitrile:water and filtered with a 2 μ m syringe filter. Analyte separation was performed with an Atlantis Premier ethylene-bridged hybrid Z-HILIC column (Waters Corp.) on a 1260 Infinity II LC system coupled to a 6460 Triple Quadrupole MS (TQMS; Agilent Technologies Inc.) in positive electrospray ionization (ESI) using Jet Stream ESI source. Data acquisition and analysis were performed with MassHunter workstation (Agilent Technologies Inc.). A U-¹³C, ¹⁵N labeled canonical AA mix (MSK-NCAA-1; Cambridge Isotope Laboratories Inc.) was used as an internal standard for blanks, standards, and samples for quantification and normalization as well as for retention time confirmation. Recovery of the internal standard was $\geq 80\%$ of the average standard curve for each AA. System blank, double blank, and blank samples were run before the standard as well as after every 10 unknown samples to prevent contamination. Quality control replicates of individual standard curve levels were analyzed every 20 samples. The inter-run CV was measured using a mid-point standard curve level and ranged from 0.74% (Trp) to 6.83% (Ala) across the runs performed for this experiment. For the TQMS, calibration was checked daily, and the ESI source was cleaned weekly.

Analytical Approach

Sample size estimation was performed before the study as previously described (Westhoff et al., 2024). Data were analyzed by repeated measures ANOVA using PROC MIXED (SAS v. 9.4; SAS Institute Inc.) with the fixed effects of treatment, time, parity (2 vs. ≥ 3), and the interaction of treatment and time. Data were analyzed according to the dietary strategy to show the change in AA and tissue mobilization over time. All models included baseline covariate measurements, the random effect of enrollment block, and repeated effect of time with a heterogeneous compound symmetry covariance structure. Model assumptions of normality and homoscedasticity of the residuals were visually assessed. An F-test was generated to determine the effect of treatment at each time point. When a treatment effect was identified, Tukey's post hoc test was used to adjust for multiple comparisons. Data are reported as least squares means $\pm$ standard error of the mean and significance was declared when $P \leq 0.05$.

RESULTS

A detailed description of the study animals, diets, DMI, milk production and composition, and metabolic indicators were reported in our companion paper (Westhoff et al., 2024). In brief, cows fed the prepartum H diet had a

greater absolute prepartum DMI (14.0 ± 0.3 vs. 13.5 ± 0.3 kg), but treatment did not affect absolute DMI postpartum or DMI as a percent of BW pre- or postpartum. From 1 to 21 DIM, ECM was greater in HH (53.6 ± 1.1 kg) compared with CC (46.4 ± 1.1 kg) and HC (45.2 ± 1.1 kg) and ECM was greater in CH (50.3 ± 1.1 kg) compared with HC, respectively. From 22 to 42 DIM, ECM was elevated in CH (58.3 ± 1.1 kg) and HH (58.0 ± 1.1 kg) compared with CC (53.1 ± 1.1 kg) and HC (53.1 ± 1.1 kg), respectively. An abbreviated nutrient and predicted AA composition of the diets are shown in Table 1. Estimated MP supply was 1,079, 1,175, and 1,603 g/d for the prepartum far-off, C, and H diets, respectively. Estimated MP supply for the postpartum C, H, and common lactation diets was 2,044, 2,685, and 2,956 g/d, respectively.

Markers of tissue mobilization

Markers of tissue mobilization are in Figure 1. Longissimus dorsi muscle diameter did not change from -14 to -7 d before expected parturition ($P = 0.99$) but decreased from -7 to 40 d relative to calving ($P < 0.01$). Longissimus dorsi muscle diameter was not affected by treatment from -14 to 21 d relative to calving but was greater in CH compared with HH at 40 DIM (37.0 ± 0.8 vs. 33.8

± 0.8 mm; $P = 0.02$), respectively. Backfat thickness increased from -14 to -7 d before expected calving ($P < 0.01$) and decreased from -7 to 40 d relative to calving ($P < 0.01$). Treatment did not affect backfat thickness ($P = 0.13$). Concentrations of creatinine were higher in CC compared with HC ($P = 0.02$) at -13 d relative to calving, and higher in CC compared with HC ($P < 0.01$) and HH ($P = 0.04$) at -6 d relative to calving, but treatment did not affect creatinine concentrations postpartum ($P \geq 0.13$). Concentrations of 3MH and the ratio of 3MH to creatinine were not affected by treatment ($P \geq 0.27$).

MP, ME, N excretion

The model estimated balance of MP and ME with and without the contribution of tissue synthesis or mobilization, determined by BCS change, are in Figure 2. Model predicted urinary and fecal N excretion are in Figure 3.

Plasma Amino Acids

Essential amino acids. A comprehensive overview of all EAA are shown in Table 2 and Figure 4. To show time dependent changes in plasma AA concentrations in the transition period, data were analyzed according to

Table 1. Nutrient composition of diets

Component ² (mean $\pm$ SD)	Treatment diets ¹					
	Far-off	Prepartum		Postpartum		Lactation ³
		Control	High	Control	High	
aNDF, % of DM	45.6 ± 3.5	43.4 ± 1.0	41.3 ± 0.4	33.3 ± 0.5	32.0 ± 0.5	27.6 ± 0.8
Starch, % of DM	15.0 ± 3.8	21.8 ± 0.7	20.9 ± 0.9	24.7 ± 0.7	24.7 ± 0.9	32.1 ± 1.3
CP, % of DM	13.3 ± 1.8	11.5 ± 0.4	14.6 ± 1.0	14.5 ± 0.3	15.9 ± 0.5	15.4 ± 0.6
Fat, % of DM	3.4 ± 0.3	3.4 ± 0.1	2.9 ± 0.1	4.9 ± 0.1	4.8 ± 0.1	4.2 ± 0.2
Estimated Mcal of ME/kg DM ⁴	2.33	2.09	2.13	2.65	2.66	2.69
Estimated MP, g/kg of DM ⁴	81	87	115	101	127	113
Estimated EAA, g/d ⁴	541	600	784	1,016	1,296	1,417
Estimated NEAA, g/d ⁴	538	575	819	1,028	1,389	1,539
Grams of AA/Mcal ME ⁴						
Methionine	0.85	1.24	1.23	1.18	1.20	1.01
Lysine	2.68	3.88	3.84	3.27	3.31	2.88
Arginine	2.19	2.58	3.42	2.28	2.95	2.60
Threonine	1.80	2.05	2.54	1.82	2.22	2.00
Leucine	2.67	3.18	4.46	2.94	4.01	3.39
Isoleucine	1.86	2.06	2.45	1.84	2.14	2.01
Valine	2.07	2.44	3.15	2.14	2.74	2.36
Histidine	0.92	1.18	1.67	1.02	1.45	1.16
Phenylalanine	1.74	2.06	2.78	1.84	2.45	2.12
Tryptophan	0.52	0.60	0.68	0.53	0.58	0.57

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 0 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP.

²Chemical composition represents the mean $\pm$ SD of six 4-wk composite samples.

³From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g of estimated MP/kg DM.

⁴Estimated in AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) using average dry matter intake for each diet.

dietary treatment. Cows fed the H prepartum diet had elevated Val and Leu concentrations as well as lower Met concentrations compared with cows fed the C prepartum diet at both -13 and -6 d relative to calving ($P \leq 0.05$). Isoleucine and Phe concentrations were greater in HH compared with CC ($P \leq 0.01$) and CH ($P \leq 0.02$) at -13 d relative to calving. At -13 d relative to calving, Arg concentration was greater in HH compared with CH ($P < 0.01$) and CC ($P < 0.01$) and Arg concentration in HC was greater than CC ($P = 0.04$), respectively. Further, Arg concentration was greater in HC compared with CC at -6 d relative to calving ($P = 0.05$). Total EAA concentration was greater at -13 d relative to calving in HH ($1,066.3 \pm 42.8 \mu\text{M}$) compared with CC ($823.4 \pm 42.9 \mu\text{M}$; $P < 0.01$) and CH ($846.3 \pm 42.9 \mu\text{M}$; $P < 0.01$) and elevated at -6 d relative to calving in HC ($958.5 \pm 35.9 \mu\text{M}$) compared with CC ($829.5 \pm 36.0 \mu\text{M}$; $P = 0.03$), respectively.

During the postpartum period, cows fed the H postpartum diet had elevated Val, Leu, and total EAA concentrations as well as lower Met concentrations compared with cows fed the C postpartum diet at 6 and 20 DIM ($P < 0.01$). Isoleucine concentration was greater in HH compared with CC ($P = 0.03$) and HC ($P = 0.03$) at 6 DIM and greater in CH compared with CC ($P = 0.04$) and HC ($P < 0.01$) at 20 DIM, respectively. At 20 DIM, Phe concentration was higher in CH and HH compared with HC ($P \leq 0.01$). Cows fed CH had elevated concentrations

of Arg compare with CC at 20 DIM ($P < 0.01$) and Arg concentration at 40 DIM was greater in HC compared with HH ($P = 0.02$), respectively. Histidine, Lys, and Trp concentrations were greater in HC compared with HH at 40 DIM ($P \leq 0.02$).

Non-essential amino acids and AA derivatives. Treatment differences for plasma concentrations of NEAA and AA derivatives are in Table 2 and Figure 5. Cows fed the H prepartum diet had lower concentrations of Ala at -6 d relative to calving compared with cows fed the C prepartum diet ($P \leq 0.03$). Hydroxy-proline was greater at -6 d relative to calving in CC compared with HC and HH ($P = 0.02$). Proline concentrations were higher at -13 d relative to calving in HH compared with CH ($P = 0.04$). Cows fed HC had lower concentrations of Ser at -6 d relative to calving compared with CH ($P = 0.03$). Total NEAA concentrations were greater in CH ($1,202 \pm 41 \mu\text{M}$) compared with HC ($993 \pm 41 \mu\text{M}$; $P < 0.01$) and HH ($1,023 \pm 41 \mu\text{M}$; $P = 0.01$) at -6 d relative to calving.

At 20 DIM, Glu concentration was greater in HC compared with CC ($P = 0.03$) and CH ($P < 0.01$) and Ala concentration was greater in HC compared with CH ($P = 0.03$). Compared with cows fed CC, Gln concentration was elevated in cows fed HH ($P = 0.01$) and Pro concentration was higher in cows fed CH and HH ($P \leq 0.04$) at 20 DIM, respectively. Cows fed HH had elevated Tyr concentrations at 6 DIM ($P = 0.04$) and lower Ser

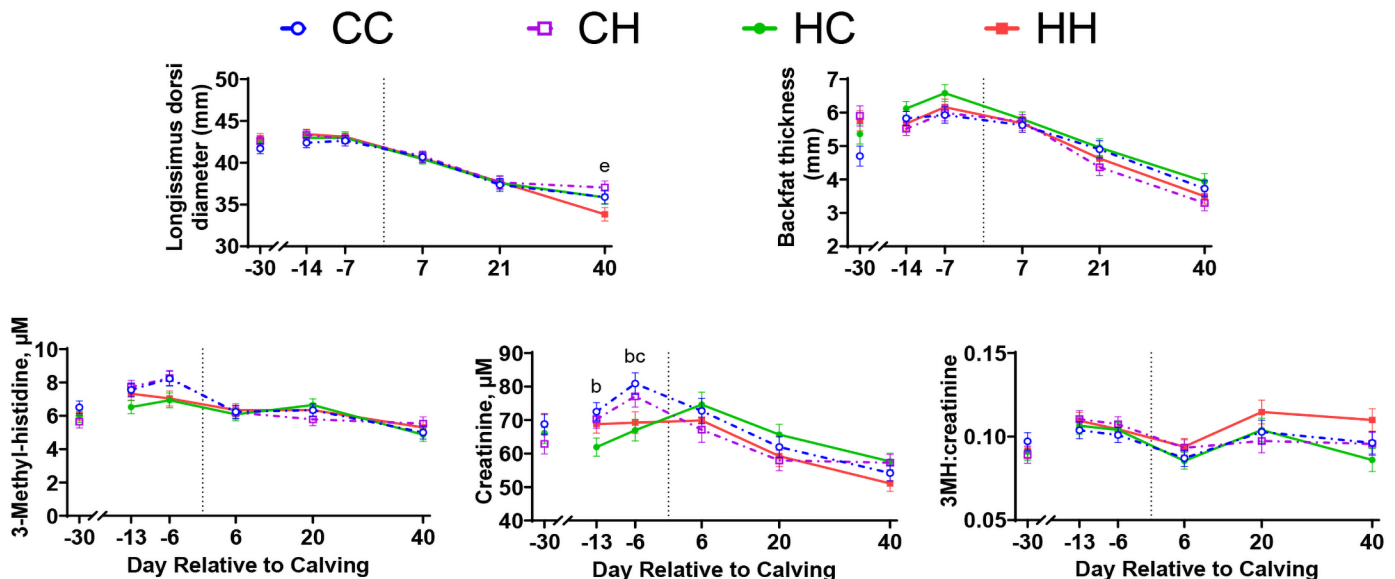


Figure 1. Effect of increasing MP supply in the prepartum, postpartum, or both diets on ultrasonographic measurement of the longissimus dorsi muscle diameter and backfat thickness as well as plasma concentrations of 3-methylhistidine, creatinine, and the ratio of 3-methylhistidine to creatinine. Prepartum diets were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg. Letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HC, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH.

concentrations at 40 DIM ($P = 0.04$) compared with cows fed HC, respectively.

DISCUSSION

Markers of tissue mobilization

Mobilization of lean muscle and adipose tissue is an inherent characteristic of the transition period as a

homeorhetic response to the negative nutrient balance (Zachut and Moallem, 2017; Peiter et al., 2023; Sadri et al., 2023). Estimates by Komaragiri and Erdman (1997) suggest that cows mobilize 21 kg of protein and 54 kg of adipose tissue from -2 to 5 wk relative to calving. However, substantial individual cow variation was observed for the duration and extent of tissue mobilization (Tamminga et al., 1997; Van der Drift et al., 2012) and

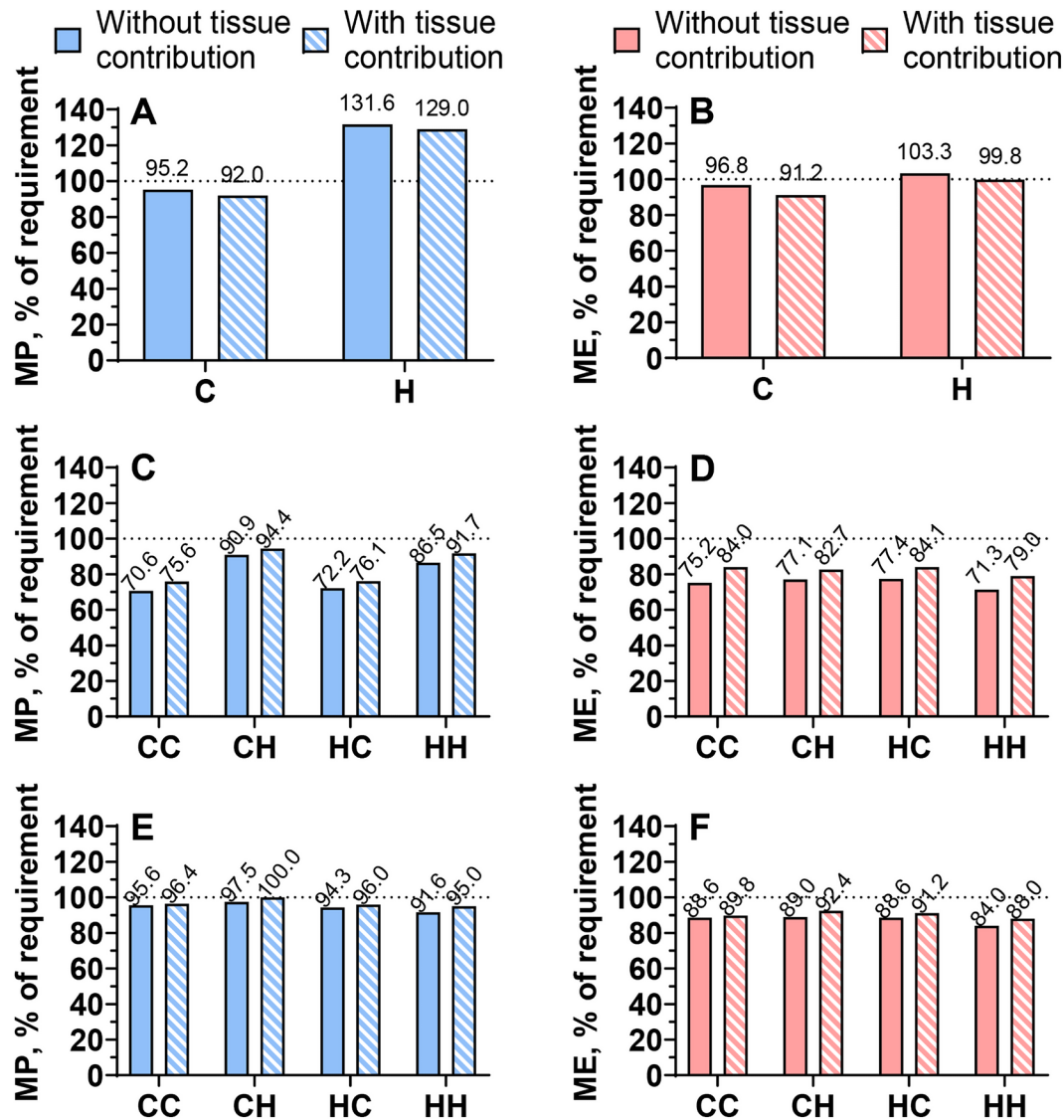


Figure 2. Estimated MP and metabolizable energy (ME) as a percentage of requirement supplied by the diet as well as when accounting for tissue synthesis or mobilization during the prepartum period (A; B), from 1 to 21 (C; D), and 22 to 42 DIM (E; F). Estimations were determined using AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) using the average DMI, BW, BCS, calf birth weight, and days carried calf for prepartum estimates and average DMI, BW, BCS, DIM, milk yield, and milk composition for postpartum estimates. The change in BCS was included during each period to describe the contribution of tissue synthesis or mobilization. Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.

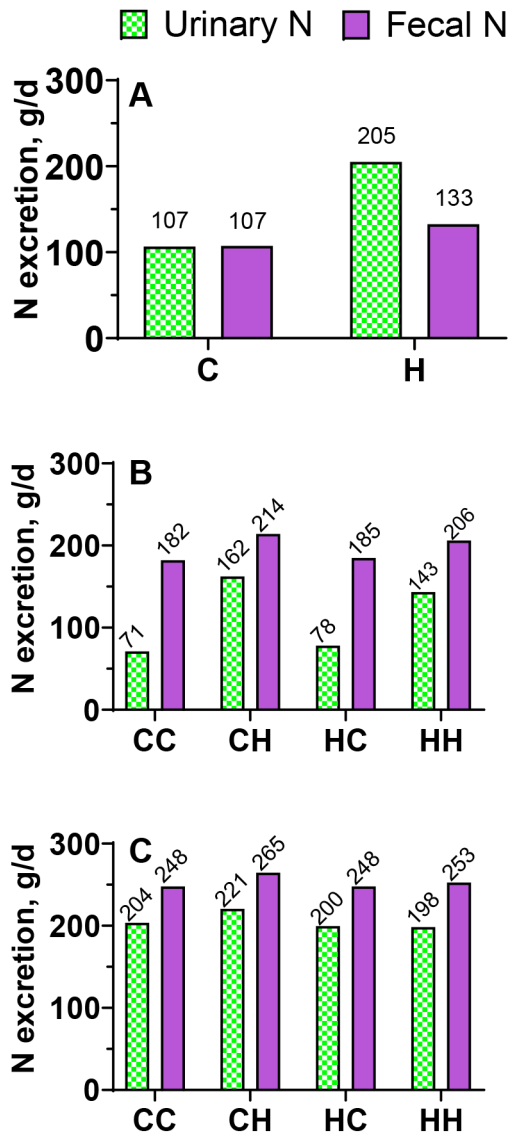


Figure 3. Estimated urinary and fecal nitrogen (N) excretion during the prepartum period (A), from 1 to 21 (B), and 22 to 42 DIM (C). Estimations were determined using AMTS.Cattle.Professional v. 4.17.0.0 (AMTS LLC) using the average DMI, BW, calf birth weight, days carried calf and change in BCS for prepartum estimates and average DMI, BW, DIM, milk yield, milk composition and change in BCS for postpartum estimates. Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal ME, respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 1 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.

this variation might, in part, be explained by differences in intake, milk production, and diet formulation. Increasing the MP or CP supply during early lactation has pre-

viously been shown to decrease plasma concentrations of 3MH (Sawada et al., 2013; Carder and Weiss, 2017; Tebbe and Weiss, 2021; Zang et al., 2021), suggesting a lower degree of lean muscle mobilization. However, when casein was infused into the abomasum at a rate designed to eliminate the protein deficit until 29 DIM, 3MH concentration did not differ compared with cows infused with water, but casein infusion increased concentrations of fatty acids at 4 DIM suggestive of greater adipose tissue mobilization (Larsen et al., 2014). In agreement, Ørskov et al. (1977) observed a linear increase in milk production and a decrease in estimated energy balance in early lactation cows abomasally infused with 0, 250, 500, or 750 g/d of casein.

Contrary to some of the above-mentioned findings, we found that indicators of tissue mobilization, including the ratio of 3MH to creatinine used as an indicator to normalize the rate of muscle mobilization to the muscle mass, as well as ultrasonographic measurement of the longissimus dorsi and backfat were not affected by MP feeding strategy during the immediate pre- and postpartum periods. Given the interrelationship between AA and ME, adjustment in the rate of tissue mobilization, as observed in cows included in the aforementioned studies, suggests an imbalance of AA and ME. This concept was demonstrated in a study by Whitelaw et al. (1986) where milk production linearly increased and protein mobilization linearly decreased when cows were abomasally infused with casein at 0, 200, 400, and 600 g/d. Concurrently, adipose tissue mobilization increased 221 and 238% when cows were infused 200 and 400 compared with 0 g of casein/d, but fat mobilization numerically decreased 21% when cows were infused 600 compared with 400 g/d (Whitelaw et al., 1986). The results shown in Whitelaw et al. (1986) are reflective of the metabolic flexibility to use excess AA when energy substrates, such as glucose, are limited. As a result of increased MP supply in the current study, cows fed the H postpartum diet produced a greater amount of milk and had an elevated demand for energy and AA compared with cows fed the C postpartum diet. While we did not observe treatment differences for indicators of adipose or lean muscle mobilization, increased concentrations of plasma urea nitrogen and BHB in cows fed the H postpartum diet (Westhoff et al., 2024) suggest that catabolism of excess AA as well as ketones were in part used to meet their energetic needs.

Plasma Amino Acids

Concentrations of plasma AA were comparable with those reported previously for the dynamic transition period (Meijer et al., 1995; Zhou et al., 2016) and reflect the balance of AA supply and utilization. Tebbe and Weiss (2021a) previously showed that increasing the MP

Table 2. Effect of increasing MP supply in the prepartum, postpartum, or both diets on plasma AA concentrations

AA, μM^3	Treatment ¹				P-value ²			
	CC (n = 15)	CH (n = 15)	HC (n = 15)	HH (n = 15)	Trt	Time	Parity	Trt x time
EAA								
Arginine	50.7 $\pm$ 2.0 ^b	55.9 $\pm$ 2.0 ^{ab}	58.8 $\pm$ 2.0 ^a	57.2 $\pm$ 2.0 ^{ab}	0.01	<0.01	0.38	<0.01
Histidine	47.7 $\pm$ 1.5	51.4 $\pm$ 1.5	51.2 $\pm$ 1.5	50.3 $\pm$ 1.5	0.19	<0.01	0.08	0.05
Isoleucine	93.6 $\pm$ 3.7 ^b	108.4 $\pm$ 3.7 ^a	98.1 $\pm$ 3.7 ^{ab}	109.3 $\pm$ 3.6 ^a	<0.01	<0.01	0.21	<0.01
Leucine	116.5 $\pm$ 4.7 ^c	152.7 $\pm$ 4.7 ^{ab}	140.7 $\pm$ 4.6 ^b	162.7 $\pm$ 4.6 ^a	<0.01	<0.01	0.86	<0.01
Lysine	70.6 $\pm$ 2.4	76.7 $\pm$ 2.5	76.6 $\pm$ 2.5	74.5 $\pm$ 2.4	0.21	<0.01	0.48	0.27
Methionine	33.1 $\pm$ 1.0 ^a	24.9 $\pm$ 1.0 ^c	29.1 $\pm$ 1.0 ^b	22.5 $\pm$ 0.9 ^c	<0.01	<0.01	<0.01	<0.01
Phenylalanine	40.0 $\pm$ 1.2 ^b	43.5 $\pm$ 1.2 ^{ab}	41.4 $\pm$ 1.2 ^{ab}	44.4 $\pm$ 1.2 ^a	<0.01	<0.01	0.09	<0.01
Threonine	91.5 $\pm$ 3.8	87.8 $\pm$ 3.7	94.4 $\pm$ 3.7	90.0 $\pm$ 3.7	0.60	<0.01	0.20	0.69
Tryptophan	26.5 $\pm$ 0.9	27.3 $\pm$ 0.9	27.9 $\pm$ 1.0	26.9 $\pm$ 0.9	0.69	<0.01	0.23	0.11
Valine	187.4 $\pm$ 7.9 ^c	248.3 $\pm$ 7.9 ^{ab}	224.3 $\pm$ 7.9 ^b	259.1 $\pm$ 7.8 ^a	<0.01	<0.01	0.14	<0.01
BCAA ⁴	397.7 $\pm$ 15.2 ^c	502.6 $\pm$ 15.1 ^{ab}	462.8 $\pm$ 15.2 ^b	531.5 $\pm$ 15.0 ^a	<0.01	<0.01	0.26	<0.01
EAA ⁵	757.4 $\pm$ 23.5 ^b	877.6 $\pm$ 23.5 ^a	843.0 $\pm$ 23.4 ^a	895.1 $\pm$ 23.2 ^a	<0.01	<0.01	0.19	<0.01
NEAA								
Alanine	231.3 $\pm$ 7.3	237.4 $\pm$ 7.2	238.4 $\pm$ 7.3	219.2 $\pm$ 7.2	0.10	<0.01	0.02	<0.01
Asparagine	59.3 $\pm$ 2.5	62.4 $\pm$ 2.5	63.0 $\pm$ 2.5	66.5 $\pm$ 2.5	0.22	<0.01	0.05	0.08
Glutamic Acid	28.0 $\pm$ 1.3	30.1 $\pm$ 1.3	28.8 $\pm$ 1.3	27.8 $\pm$ 1.3	0.51	<0.01	0.59	<0.01
Glutamine	203.5 $\pm$ 7.4 ^b	232.5 $\pm$ 7.1 ^a	223.2 $\pm$ 7.1 ^{ab}	226.2 $\pm$ 7.0 ^a	<0.01	<0.01	0.33	0.10
Glycine	427.9 $\pm$ 25.6	461.1 $\pm$ 25.6	424.9 $\pm$ 25.7	436.0 $\pm$ 25.5	0.72	<0.01	0.30	0.69
Proline	72.3 $\pm$ 2.5	77.5 $\pm$ 2.5	76.8 $\pm$ 2.5	78.6 $\pm$ 2.5	0.25	<0.01	0.05	0.02
Serine	96.4 $\pm$ 3.8	98.3 $\pm$ 3.8	93.7 $\pm$ 3.8	94.5 $\pm$ 3.8	0.76	<0.01	0.04	0.06
Tyrosine	35.6 $\pm$ 1.4	39.9 $\pm$ 1.4	37.7 $\pm$ 1.4	38.0 $\pm$ 1.4	0.08	<0.01	0.09	0.02
NEAA ⁶	1,145 $\pm$ 34	1,244 $\pm$ 34	1,184 $\pm$ 34	1,179 $\pm$ 34	0.20	<0.01	0.02	0.13
1-Methyl-Histidine	9.0 $\pm$ 0.5 ^a	7.7 $\pm$ 0.5 ^{ab}	7.5 $\pm$ 1.5 ^{ab}	7.0 $\pm$ 0.5 ^b	0.03	<0.01	0.28	<0.01
3-Methyl-Histidine	6.7 $\pm$ 0.3	6.7 $\pm$ 0.3	6.2 $\pm$ 0.3	6.5 $\pm$ 0.3	0.51	<0.01	0.50	0.02
Creatinine	68.5 $\pm$ 2.2	65.9 $\pm$ 2.2	65.4 $\pm$ 2.2	63.6 $\pm$ 2.2	0.39	<0.01	0.01	<0.01
3MH:creatinine	0.098 $\pm$ 0.004	0.101 $\pm$ 0.004	0.097 $\pm$ 0.004	0.107 $\pm$ 0.004	0.27	<0.01	<0.01	0.61
Carnosine	11.5 $\pm$ 0.5	12.4 $\pm$ 0.5	12.3 $\pm$ 0.5	12.7 $\pm$ 0.5	0.23	<0.01	0.72	0.15
Citrulline	57.8 $\pm$ 2.3 ^a	66.3 $\pm$ 2.3 ^b	66.1 $\pm$ 2.3 ^b	68.8 $\pm$ 2.3 ^b	<0.01	<0.01	0.60	<0.01
Hydroxyproline	57.1 $\pm$ 1.9	57.0 $\pm$ 1.9	55.3 $\pm$ 1.9	55.3 $\pm$ 1.9	0.75	<0.01	0.50	<0.01
Ornithine	31.2 $\pm$ 2.0	31.8 $\pm$ 2.0	37.0 $\pm$ 2.0	36.8 $\pm$ 2.0	0.03	<0.01	0.06	0.02

^{a-c} Least squares means with different superscripts differ ($P \leq 0.05$; Tukey's test).

¹Prepartum diets, fed starting at 28 d before expected calving, were formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. Postpartum diets, fed from 0 to 21 DIM, were formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet that contained Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supplied 113 g MP/kg.

²Fixed effects of treatment (trt), time (−30, −13, −6, 6, 20, and 40 d relative to calving), parity (2 vs. ≥ 3), and the interaction of treatment x week.

³Reported as LSM $\pm$ SEM.

⁴Sum of Ile, Leu, and Val.

⁵Sum of Arg, His, Ile, Leu, Lys, Met, Phe, Thr, Try, and Val.

⁶Sum of Ala, Asn, Glu, Gln, Gly, Pro, Ser, and Tyr.

supply during early lactation affected plasma concentrations of several EAA and NEAA. Here, we also observed treatment differences for the concentration of several plasma AA including BCAA that were elevated in cows fed the H diet pre- and postpartum. Branched-chain AA are integral for milk synthesis accounting for 50% of the EAA secreted in milk. In addition, BCAA are involved in intracellular pathways important for regulation of protein synthesis, including the mammalian target of rapamycin (mTOR) in bovine mammary tissue (Appuhamy et al., 2012). They are also taken up by the mammary gland in excess of the requirement for milk protein to supply intermediates for the tricarboxylic acid cycle and for synthesis of NEAA (Mackle et al., 1999; Lei et al., 2012;

Larsen et al., 2015). A study by Larsen et al. (2014) demonstrated that increasing the MP supply using abomasal casein infusion in early lactation cows did not alter the ratio of BCAA uptake to output in the mammary gland. In theory, an elevated availability of circulating BCAA in cows fed the H postpartum diets could, in part, contribute to the regulation as well as energetic and AA demands of the mammary gland to support a greater milk production as observed in these treatments. In support of this theory, increased yields of milk, milk fat, and lactose were observed from 1 to 21 DIM in cows that received abomasal infusion of BCAA (Gallagher et al., 2024).

Methionine and Lys are considered the first-limiting AA for dairy cattle (Schwab et al., 1976). Because the

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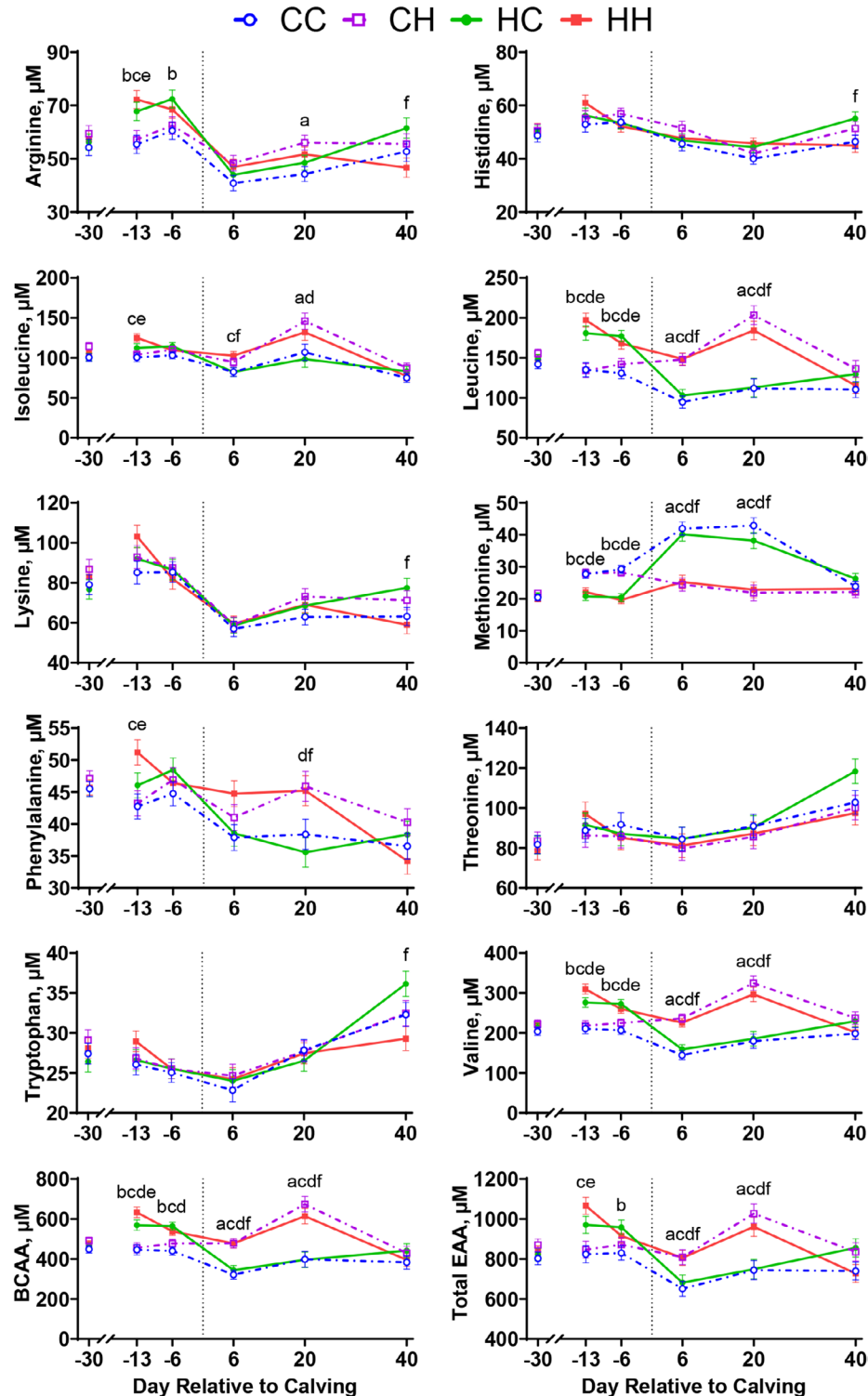


Figure 4. Plasma concentrations of EAA from cows ($n = 60$) fed a prepartum TMR, starting at 28 d before expected calving, formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply either a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. A postpartum TMR was fed from 0 to 21 DIM and formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet formulated to contain Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supply 113 g MP/kg DM. Letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HC, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH. BCAA = sum of Ile, Leu, and Val. EAA = sum of Arg, His, Ile, Leu, Lys, Met, Phe, Thr, Try, and Val. P -values for all main effects are in Table 2.

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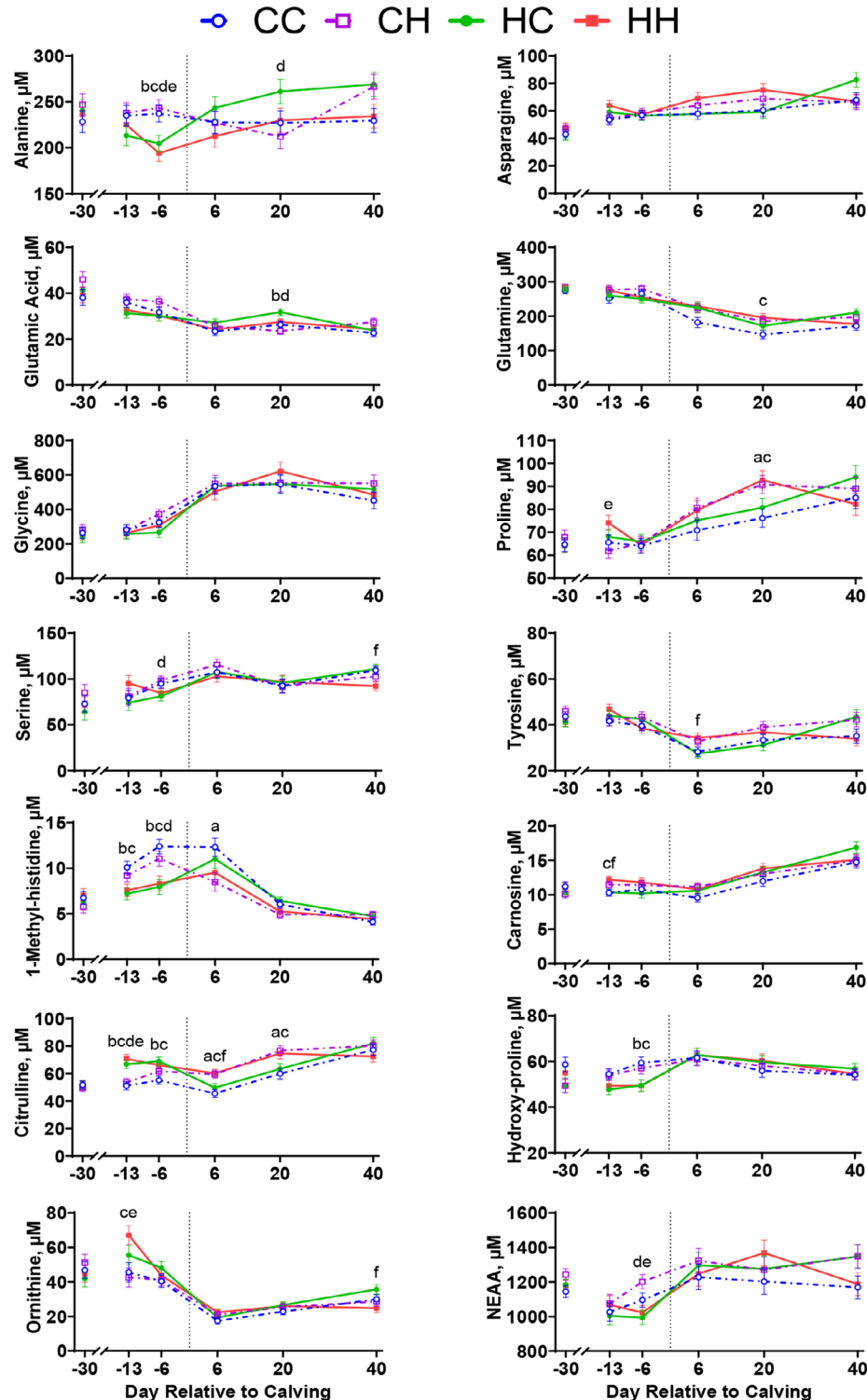


Figure 5. Plasma concentrations of NEAA from cows ($n = 60$) fed a prepartum TMR, starting at 28 d before expected calving, formulated to contain Met and Lys at 1.24 and 3.84 g/Mcal metabolizable energy (ME), respectively and supply either a control (C; 85 g MP/kg DM) or high (H; 113 g MP/kg DM) level of estimated MP. A postpartum TMR was fed from 0 to 21 DIM and formulated to contain Met and Lys at 1.15 and 3.16 g/Mcal ME, respectively and supply control (C; 104 g MP/kg DM) or high (H; 131 g MP/kg DM) level of estimated MP. The combination of a pre- and postpartum diet resulted in treatments: 1) CC, 2) CH, 3) HC, 4) HH. From 22 to 42 DIM, cows were fed a lactation diet formulated to contain Met and Lys at 1.01 and 2.88 g/Mcal ME, respectively and supply 113 g MP/kg DM. Letters denote differences ($P \leq 0.05$; Tukey's test) between treatments at individual time point. a = CC vs. CH, b = CC vs. HC, c = CC vs. HH, d = CH vs. HC, e = CH vs. HH, f = HC vs. HH. NEAA = sum of Ala, Asn, Glu, Gln, Gly, Pro, Ser, and Tyr. P -values for all main effects are in Table 2.

importance of Met and Lys on animal productivity during the transition period has been firmly established (Socha et al., 2005; Toledo et al., 2021; Elsaadawy et al., 2022), treatment diets were formulated to balance for these AA per Mcal of ME in the current study. While plasma concentrations of Lys did not differ by treatment, we observed a treatment effect for Met concentration such that cows fed the H diets had lower concentrations of Met in the pre- and postpartum periods. As a group 1 AA, the uptake of Met by the mammary gland was closely reflected by the output in milk protein (Lapierre et al., 2012). During the first 21 d of lactation, protein yield was 90 to 150 g/d greater in cows fed the H compared the C postpartum diet (Westhoff et al., 2024); thus these cows secreted an additional 1.6 to 2.7 g/d of Met in milk (Khan et al., 2019) that equates to approximately 2.5 to 4.0% of daily Met intake. Beyond its role in milk protein synthesis, Met has been credited for its involvement in several biological functions during the transition period. Most authors to date have designed studies to isolate the effects of Met supplementation (Osorio et al., 2014; Chandler and White, 2017; Thanh et al., 2023). Therefore, our understanding of Met metabolism when the supply is kept constant, but other AA are increased, as done in the current study, remains limited. In work by Larsen et al. (2015) and Tucker et al. (2021), infusion of AA or milk protein increased the fractional hepatic removal of Met at 5 DIM but did not affect hepatic Met oxidation compared with control cows. These data suggest that the increased Met uptake by the liver when cows were supplemented with MP was diverted toward metabolic functions rather than catabolism. As such, our data confirms that an increased milk protein yield, in part, contributed to the lower plasma concentrations of Met and an increased supply of other AA likely synergistically increasing hepatic Met utilization, but we are not able to rule out the possibility of a diet difference in the flow of Met to the duodenum that could have affected plasma concentrations.

The nadir in concentrations of total EAA during early lactation observed in the current study reinforced observations by previous authors (Meijer et al., 1995; Doepel et al., 2002; Kuhla et al., 2011) and aligns with the nadir in protein balance at 7 DIM calculated by Bell et al. (2000). In tandem with a lower DMI and an increase in the net hepatic uptake of group 1 AA that was proportional to portal absorption, EAA removal efficiency by the mammary gland would have increased to support the output of milk and milk protein (Lapierre et al., 2012; Larsen et al., 2015; Martineau et al., 2017) contributing to the decreased plasma concentrations during early lactation. Further, elevated plasma concentrations of NEAA, specifically Ala, Gly, and Ser, could be of great importance to hepatic gluconeogenesis (Overton et al., 1999; Larsen

and Kristensen, 2012; 2013). Because all EAA except Met and Lys were increased in H compared with C diets, elevated concentrations of milk and plasma urea nitrogen in cows fed the H diets in the current study (Westhoff et al., 2024) suggest increased catabolism of excess AA that could hypothetically spare glucose for milk synthesis in the mammary gland.

CONCLUSIONS

Our results show that altering the periparturient MP supply, with diets formulated to contain an equal amount of Met and Lys on an ME basis, did not affect indicators of adipose and muscle tissue mobilization. Treatment affected concentrations of circulating EAA such that cows fed an elevated MP supply postpartum had greater concentrations of total EAA at 6 and 20 DIM. Plasma concentrations of total NEAA were affected by treatment at -6 d relative to calving but did not differ from 6 to 40 DIM. Our findings suggest that altering the pre- and postpartum MP supply affected plasma AA concentrations, but indicators of tissue mobilization did not differ during the treatment period. Overall, our results indicate that increasing MP supply influences nutrient partitioning toward the mammary gland.

ACKNOWLEDGMENTS

This study was supported in part by the Agriculture and Food Research Initiative competitive grant number 2019-67015-29834 from the USDA National Institute of Food and Agriculture (Washington, DC) and Purina Animal Nutrition (Gray Summit, MO). Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the view of the USDA or Purina Animal Nutrition. We thank Jeff Tikofsky (Zinpro Corporation) for the expertise in diet formulation and Charlene Ryan, Amy Stabell, and Rebecca Coombe (Cornell University) for technical assistance during the study. The authors acknowledge the staff at the Cornell University Ruminant Center for care of the animals. The authors have not stated any conflicts of interests.

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